SYLLABUS

Full Syllabus of all five selected Courses.

1. **Introduction to Machine Learning**
2. **Data Structures**
3. **Operating Systems**
4. **Database Management Systems**
5. **Foundation of Data Science**

Course Contents and Lecture Schedule

No	Topic	No. of Lectures
1	**INTRODUCTION TO MACHINE LEARNING**	
1.1	Mock Test on Module 1, Module 2 and Module 3	1 hour
1.2	Mock Test on Module 4 and Module 5	1 hour
1.3	Feedback and Remedial class	1 hour
2	**DATA STRUCTURES**	
2.1	Mock Test on Module 1, Module 2 and Module 3	1 hour
2.2	Mock Test on Module 4 and Module 5	1 hour
2.3	Feedback and Remedial class	1 hour
3	**OPERATING SYSTEMS**	
3.1	Mock Test on Module 1 and Module 2	1 hour
3.2	Mock Test on Module 3, Module 4 and Module 5	1 hour
3.3	Feedback and Remedial class	1 hour
4	**DATABASE MANAGEMENT SYSTEMS**	
4.1	Mock Test on Module 1, Module 2 and Module 3	1 hour
4.2	Mock Test on Module 4 and Module 5	1 hour

4.3	Feedback and Remedial class	1 hour
5	**FOUNDATIONS OF DATA SCIENCE**	
5.1	Mock Test on Module 1, Module 2 and Module 3	1 hour
5.2	Mock Test on Module 4 and Module 5	1 hour
5.3	Feedback and Remedial class	1 hour

Model Question Paper

QP CODE:

Reg No: _______________

Name: _________________ **PAGES :7**

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

SIXTH SEMESTER B.TECH DEGREE EXAMINATION, MONTH & YEAR

Course Code: ADT308

Course Name: Comprehensive Course Work

Max. Marks: 50 **Duration: 1 Hour**

Objective type questions with multiple choices. Mark one correct answer for each question. Each Question Carries 1 Mark

1. Application of machine learning methods to large databases is called

   (A) Data Mining (B) Artificial Intelligence

   (C) Big Data Computing (D) Internet of Things

2. If machine learning model output involves target variable, then that model is called as

   (A) Descriptive Model (B) Predictive Model

   (C) Reinforcement Learning (D) All of the above

3. In what type of learning labelled training data is used

   (A) Unsupervised Learning (B) Supervised Learning

   (C) Reinforcement Learning (D) Active Learning

4. In following type of feature selection method we start with empty feature set

(A) Forward Feature Selection (B) Backward Feature Selection

(C) Both A and B (D) None of the above

5. Which of the following is the best machine learning method?

(A) Scalable (B) Accuracy

(C) Fast (D) All of the above

6. Data used to build a data mining model.

(A) Training data (B) Validation data

(C) Test data (D) Hidden data

7. You are given reviews of few netflix series marked as positive, negative and neutral. Classifying reviews of a new netflix series is an example of

(A) Supervised learning (B) Unsupervised learning

(C) Semisupervised learning (D) Reinforcement learning

8. Following are the types of supervised learning

(A) Classification (B) Regression

(C) subgroup discovery (D) all of the above

9. The output of training process in machine learning is

(A) machine learning model (B) machine learning algorithm

(C) null (D) accuracy

10. PCA is

(A) forward feature selection (B) backward feature selection

(C) feature extraction (D) all of the above

11. Consider the following sequence of operations on an empty stack.
push(22); push(43); pop(); push(55); push(12); s=pop();
Consider the following sequence of operations on an empty queue.
enqueue(32);enqueue(27); dequeue(); enqueue(38); enqueue(12); q=dequeue();
The value of s+q is ___________
(A) 44 (B) 54 (C) 39 (D) 70

12. A B-tree of order (degree)5 and of height 3 will have a minimum of ___ keys.
A. 624
B. 249
C. 124
D. 250

13. Construct a binary search tree by inserting 8, 6, 12, 3, 10, 9 one after another. To make the resulting tree as AVL tree which of the following is required?
(A) One right rotation only
(B) One left rotation followed by two right rotations
(C) One left rotation and one right rotation
(D) The resulting tree itself is AVL

14. In a complete 4-ary tree, every internal node has exactly 4 children or no child. The number of leaves in such a tree with 6 internal nodes is:
(A) 20 (B) 18 (C) 19 (D) 17

15. Select the postfix expression for the infix expression *a+b-c+d*(e/f)*.
(A) ab+c-d+e*f/ (B) ab+c-def/*+
(C) abc-+def/*+ (D) ab+c-def/*+

16. Consider a hash table of size seven, with starting index zero, and a hash function (2x + 5)mod7. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 4, 9, 6 is inserted into the table using closed hashing? Note that '_' denotes an empty location in the table.
(A) 9, _, 1, 6, _, _, 4 (B) 1, _, 6, 9, _, _, 4
(C) 4, _, 9, 6, _, _, 1 (D) 1, _, 9, 6, _, _, 4

17. **Compute the time complexity of the following function:**

```
void function(int n)
{
        int count = 0;
        for (int i=n/2; i<=n; i++)
                for (int j=1; j<=n; j = j + 2)
                        for (int k=1; k<=n; k = k * 2)
                                count++;}
```

A. $O(n^2 \log n)$
B. $O(n \log^2 n)$
C. $O(n^3)$
D. $O(n \log n^2)$

18. How many distinct binary search trees can be created out of 6 distinct keys?
(A) 7 (B) 36 (C) 140 (D) 132

19. Which tree traversal performed on a binary search tree, results in ascending order listing of the keys?
A. Pre-order
B. In-order
C. Post-order
D. Level-order

20. You are given pointers to first and last nodes of a singly linked list, which of the following operations are dependent on the length of the linked list?
    (A) Delete the first element
    (B) Insert a new element as a first element
    (C) Add a new element at the end of the list
    (D) Delete the last element of the list

21. Suppose a disk has 400 cylinders, numbered from 0 to 399. At some time the disk arm is at cylinder 58, and there is a queue of disk access requests for cylinder 66, 349, 201, 110, 38, 84, 226, 70, 86. If Shortest-Seek Time First (SSTF) is being used for scheduling the disk access, the request for cylinder 86 is serviced after servicing ____________ number of requests.
    (A) 1 (B) 2 (C)3 (D)4

22. If frame size is 4KB then a paging system with page table entry of 2 bytes can address _______ bytes of physical memory.
    (A) 2^12 (B) 2^16 (C) 2^18 (D) 2^28

23. Calculate the internal fragmentation if page size is 4KB and process size is 103KB.
    (A) 3KB (B) 4KB (C) 1KB (D) 2KB

24. Which of the following scheduling policy is likely to improve interactiveness?
    (A) FCFS (B) Round Robin
    (C) Shortest Process Next (D) Priority Based Scgeduling

25. Consider the following program

```
        Semaphore X=1, Y=0
Void A ( )                      Void B ( )
{                               {
        While (1)                       While (1)
        {                               {
        P(X);                           P(Y);
        Print'1';                       P(X);
        V(Y);                           Print'0';
        }                               V(X);
}                                       }
                                }
```

    The possible output of the program:
    (A) Any number of 0's followed by any number of 1's.
    (B) Any number of 1's followed by any number of 0's.
    (C) 0 followed by deadlock
    (D) 1 followed by deadlock

26. In a system using single processor, a new process arrives at the rate of 12 processes per minute and each such process requires 5 seconds of service time. What is the percentage of CPU utilization?

(A) 41.66 (B) 100.00 (C) 240.00 (D) 60.00

27. A system has two processes and three identical resources. Each process needs two resources to proceed. Then
(A) Deadlock is possible (B) Deadlock is not possible
(C) Starvation may be present (D) Thrashing

28. Which of the following is true with regard to Round Robin scheduling technique?
(A) Responds poorly to short process with small time quantum.
(B) Works like SJF for larger time quantum
(C) Does not use a prior knowledge of burst times of processes.
(D) Ensure that the ready queue is always of the same size.

29. Thrashing can be avoided if
(A) the pages, belonging to working set of programs, are in main memory
(B) the speed of CPU is increased
(C) the speed of I/O processor is increased
(D) none of the above

30. The circular wait condition can be prevented by
(A) using thread
(B) defining a linear ordering of resource types
(C) using pipes
(D) all of the above

31. Let E1, E2 and E3 be three entities in an E/R diagram with simple single-valued attributes. R1 and R2 are two relationships between E1 and E2, where R1 is one-to-many, R2 is many-to-many. R3 is another relationship between E2 and E3 which is many-to-many. R1, R2 and R3 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model?
(A) 3 (B) 4 (C) 5 (D) 6

32. Identify the minimal key for relational scheme R(U, V, W, X, Y, Z) with functional dependencies $F = \{U \rightarrow V, V \rightarrow W, W \rightarrow X, VX \rightarrow Z\}$
(A) UV (B) UW (C) UX (D) UY

33. It is given that: “Every student need to register one course and each course registered by many students”, what is the cardinality of the relation say “Register” from the “Student” entity to the “Course” entity in the ER diagram to implement the given requirement.
(A) M:1 relationship (B) M:N relationship
(C) 1:1 relationship (D) option (B) or(C)

34. Consider the relation branch(branch_name, assets, branch_city)
SELECT DISTINCT T.branch_name FROM branch T, branch S WHERE T.assets>L.assets AND S.branch_city = "TVM" .
Finds the names of
(A) All branches that have greater assets than all branches located in TVM.

(B) All branches that have greater assets than some branch located in TVM.
(C) The branch that has the greatest asset in TVM.
(D) Any branch that has greater asset than any branch located in TVM.

35. Consider the following relation instance, where "A" is primary Key.

A1	A2	A3	A4
1	1	1	Null
5	2	5	1
9	5	13	5
13	13	9	15

Which one of the following can be a foreign key that refers to the same relation?
(A) A2 (B) A3 (C) A4 (D) ALL

36. A relation R(ABC) is having the tuples(1,2,1),(1,2,2),(1,3,1) and (2,3,2). Which of the following functional dependencies holds well?
(A) A → BC (B) AC → B (C) AB → C (D) BC → A

37. Consider a relation R with attributes A, B, C, D and E and functional dependencies A→BC, BC→E, E→DA. What is the highest normal form that the relation satisfies?
(A) BCNF (B) 3 NF (C) 2 NF (D) 1 NF

38. For the given schedule S, find out the conflict equivalent schedule.
S : r1(x); r2(Z) ; r3(X); r1(Z); r2(Y); r3(Y);W1(X); W2(Z); W3(Y); W2(Y)
(A) T1→T2→T3 (B) T2->T1->T3
(C) T3→T1→T2 (D) Not conflict serializable

39. Specialization is __________ process.
(A) top-down (B) bottom up
(C) Both (A) and (B) (D) none of these

40. If D1, D2, .., Dn are domains in a relational model, then the relation is a table, which is a subset of
(A) D1+D2+ … +Dn (B) D1×D2× … ×Dn
(C) D1∪D2∪⋯ ∪Dn (D) D1–D2– … –Dn

41.For each value of the ___, the distribution of the dependent variable must be normal.

(A) Independent variable (B)Depended variable

(C)Intermediate variable (D)None of the mentioned above

42. Data Analytics uses ___ to get insights from data.

(A)Statistical figures (B)Numerical aspects

(C)Statistical methods (D)None of the mentioned above

43. Linear Regression is the supervised machine learning model in which the model finds the best fit ___ between the independent and dependent variable.

(A)Linear line (B)Nonlinear line

(C) Curved line (D) All of the mentioned above

44. Amongst which of the following is / are the types of Linear Regression,

(A) Simple Linear Regression (B)Multiple Linear Regression

(C)Both A and B (D)None of the mentioned above

45. Amongst which of the following is / are the true about regression analysis?

(A)Describes associations within the data

(B)Modeling relationships within the data

(C)Answering yes/no questions about the data

(D)All of the mentioned above

46. The process of quantifying data is referred to as ___.

(A)Decoding (B)Structure

(C)Enumeration (D)Coding

47. Data Analysis is a process of,

(A)Inspecting data (B)Data Cleaning

(C) Transforming of data (D)All of the mentioned above

48. Least Square Method uses ___.

(A)Linear polynomial (B)Linear regression

(C) Linear sequence (D)None of the mentioned above

49. What is a hypothesis?

(A)A statement that the researcher wants to test through the data collected in a study

(B)A research question the results will answer

(C)A theory that underpins the study

(D)A statistical method for calculating the extent to which the results could have happened by chance

50. ___ are used when we want to visually examine the relationship between two quantitative variables.

(A)Bar graph (B)Scatterplot

(C)Line graph (D)Pie chart

QNo	Ans. Key	QNo	Ans. Key	QNo	Ans. Key	QNo	Ans. Key	QNo	Ans. Key
1	(A)	11	(C)	21	(C)	31	(C)	41	(A)
2	(B)	12	(B)	22	(D)	32	(A)	42	(C)
3	(B)	13	(A)	23	(C)	33	(D)	43	(A)
4	(A)	14	(C)	24	(B)	34	(C)	44	(C)
5	(D)	15	(D)	25	(D)	35	(B)	45	(B)
6	(A)	16	(D)	26	(B)	36	(D)	46	(C)
7	(A)	17	(A)	27	(B)	37	(D)	47	(D)
8	(D)	18	(D)	28	(C)	38	(B)	48	(B)
9	(A)	19	(B)	29	(A)	39	(C)	49	(A)
10	(C)	20	(D)	30	(B)	40	(D)	50	(A)

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Sixth Semester B.Tech Degree Regular and Supplementary Examination June 2023 (2019 Scheme)

Course Code: ADT308

Course name: COMPREHENSIVE COURSE WORK

Max. Marks: 50 Duration: 1Hour

Instructions: *(1) Each question carries one mark. No negative marks for wrong answers*
(2) Total number of questions: 50
(3) All questions are to be answered. Each question will be followed by 4 possible answers of which only ONE is correct.
(4) If more than one option is chosen, it will not be considered for valuation.

1. The problem of finding hidden structure in unlabelled data is called

a) Supervised learning b) Unsupervised Learning c) Reinforcement Learning d) None

2. The ratio of correctly predicted labels and the total number of predicted labels is termed as

a) Accuracy b) Precision c) F1 score d) Confusion matrix

3. Principal Components Analysis (PCA) is used for

a) Cross Validation b) Loss estimation c) Dimension Reduction d) Performance metric

4. Support vector machine (SVM) algorithm solves

a) Entropy Maximization b) Global minima c) Local minima d) Convex optimization problem

5. Which of the following is a heuristic for k-means clustering algorithm?

a) Ward's algorithm b) Lloyd's algorithm c) Both a and b d) None

6. You are given seismic data and you want to predict next earthquake, this is an example of

a) Supervised learning b) Unsupervised Learning c) Reinforcement Learning d) Dimension reduction

7. A feature F1 can take certain value: A, B, C, D, E, & F and represents grade of students from a college. Here feature type is.

a) Nominal b) Categorical c) Boolean d) Ordinal

8. ML is a field of AI consisting of learning algorithms that

a) Improve their performance

b) At executing some task

c) Over time with experience

d) All of the above

9. The model will be trained with data in one single batch is known as

a) Batch learning b) Offline learning c) Both a and b d) None

10. Which of the following is a widely used and effective machine learning algorithm based on the idea of bagging?

a) Decision Tree b) Regression c) Classification d) Random Forest

11 Which sorting algorithm is best when list is already sorted

a) Quick sort b) Merge sort c) Insertion sort d) Selection sort

12 What would be the output after performing the following operations in a Deque

1. Insertfront(10);
2. Insertfront(20);
3. Insertrear(30);
4. Insertrear(40);
5. Deletefront();
6. Insertfront(50);
7. Deleterear();
8. Display();

a) 10,20,30 b) 50,10,30 c) 40,20,30 d) 20,30,40

13 Which of the following is a Divide and Conquer algorithm

a) Bubble b) Heap c) Selection d) Merge

14 Select the postfix expression for the infix expression *a+b-c+d*(e/f)*

a) ab+c-d+e*f/ b) ab+c-def/*+ c) abc-+def/*+ d) ab+c-def/*+

15 What is the time complexity of the binary search algorithm

a) O(n) b) O(1) c) O(log2n) d) O(n^2)

16 Which of the following algorithms are used to find the shortest path from a source node to all other nodes in a weighted graph

a) Djikstra's algorithm b) BFS c) Prims algorithm d) Kruskal's algorithm

17 Consider the following sequence of operations on an empty stack.
push(22); push(43); pop(); push(55); push(12); s=pop();
Consider the following sequence of operations on an empty queue.
enqueue(32);enqueue(27); dequeue(); enqueue(38); enqueue(12); q=dequeue();
The value of s+q is

a) 44 b) 54 c) 39 d) 70

18 What is the outcome of the prefix expression +, -, *, 3, 2, /, 8, 4, 1

a) 4 b) 5 c) 11 d) 12

19 Which of the following options is not true about the Binary Search tree

a) The value of the left child should be less than the root node

b) The value of the right child should be greater than the root node.

c) The left and right sub trees should also be a binary search tree

d) None

20 Which of these adjacency matrices represents a simple graph

a) [[1, 0, 0], [0, 1, 0], [0, 1, 1]]

b) [[1, 1, 1], [1, 1, 1], [1, 1, 1]]

c) [[0, 0, 1], [0, 0, 0], [0, 0, 1]]

d) [[0, 0, 1], [1, 0, 1], [1, 0, 0]]

21 Calculate the internal fragmentation if page size is 4KB and process size is 103KB

a) 1KB b) 2KB c) 3KB d) 4KB

22 A process which is copied from main memory to secondary memory on the basis of requirement is known as

a) Paging b) Demand paging c) Thread d) Segmentation

23 If frame size is 4KB then a paging system with page table entry of 2 bytes can address _______ bytes of physical memory.

a) 2^12 b) 2^16 c) 2^18 d) 2^28

24 Which of the following is true with regard to Round Robin scheduling technique

a) Responds poorly to short process with small time quantum

b) Works like SJF for larger time quantum

c) Does not use a prior knowledge of burst times of processes

d) Ensure that the ready queue is always of the same size.

25 FIFO scheduling is a type of

a) Non-pre-emptive scheduling

b) Pre-emptive scheduling

c) Deadline scheduling

d) None

26 For which of the following purposes in Banker's algorithm is used

a) Preventing deadlock b) Solving deadlock

c) Recovery from deadlock d) None

27 In a timeshare operating system, when the time slot assigned to a process is completed, the process switches from the current state to?

a) Suspended state b) Terminated state c) Blocked state d) Ready state

28 The operating system keeps a small table containing information about all open files called

a) File-open table b) Open directory table c) open-file table d) system table

29 The _______ program initializes all aspects of the system, from CPU registers to device controllers and the contents of main memory, and then starts the operating system.

a) bootstrap b) main c) bootloader d) rom

30 The operating system and the other processes are protected from being modified by an already running process because

a) They have a protection algorithm

b) They are in different memory spaces

c) Every address generated by the CPU is being checked against the relocation and limit registers

d) They are in different logical addresses

31 In E-R diagram an attribute is represented by

a) Circle b) Rectangle c) Square d) Ellipse

32 Which of the following SQL command is used for removing (or deleting) a relation form the database?

a) Drop b) Delete c) Roll back d) Remove

33 Which of the following can replace the below query?

SELECT name, course_id
FROM instructor, teaches
WHERE instructor_ID= teaches_ID;

a) Select name, course_id from instructor natural join teaches;

b) Select name, course_id from teaches, instructor where instructor_id=course_id;

c) Select name, course_id from instructor;

d) Select course_id from instructor join teaches;

34 Let E1, E2 and E3 be three entities in an E/R diagram with simple single-valued attributes. R1 and R2 are two relationships between E1 and E2, where R1 is one-to-many, R2 is many- to-many. R3 is another relationship between E2 and E3 which is many-to-many. R1, R2 and R3 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model

a) 3 b) 4 c) 5 d) 6

35 Find the highest normal form of a relation R(A,B,C,D,E) with Functional Dependency {B->A, A->C, BC->D, AC->BE}

a) 1NF b) 2NF c) 3NF d) BCNF

36 To select some particular columns, which of the following command is used?

a) SELECTION b) JOIN c) UNION d) PROJECTION

37 Select the definition of the correct key which is used to represent relation between two tables?

a) Primary key b) Foreign key c) Super key d) Candidate key

38 Which of the following SQL statement can be used to select the name of customers who have no

address.

a) Select Name from CUSTOMER where address IS EMPTY;

b) Select Name from CUSTOMER where address IS NULL;

c) Select Name from CUSTOMER where address=NULL;

d) All of the above

39 What do you mean by one to many relationships?

a) One class may have many teachers

b) One teacher can have many classes

c) Many classes may have many teachers

d) Many teachers may have many classes

40 In SQL which command is used to issue multiple CREATE TABLE, CREATE VIEW and GRANT statements in a single transaction.

a) Create package b) Create schema c) Create cluster d) All of the above

41 __ are used when we want to visually examine the relationship between two quantitative variable

a) Scatterplot b) Bar graph c) Pie chart d) Line graph

42 Which of the following language is used in Data science

a) R b) C c) C++ d) Ruby

43 A ________ is a structured representation of data

a) database table b) functions c) data preparation d) data frame

44 Choose a disadvantage of decision trees among the following.

a) Decision trees are prone to over fitting

b) Decision trees are robust to outliers

c) Factor analysis

d) All of the above

45 Amongst which of the following is / are the true about regression analysis

a) Describes associations within the data

b) Modelling relationships within the data

c) Answering yes/no questions about the data

d) All of the mentioned above

46 Inference engines work on the principle of?

a) Forward chaining b) Simple chaining c) Backward chaining d) Both a and c

47 How do we perform Bayesian classification when some features are missing

a) We integrate the posteriors probabilities over the missing features

b) We ignore the missing features

c) We assuming the missing values as the mean of all values

d) Drop the features completely

48 The modern conception of data science as an independent discipline is sometimes attributed to?

a) John McCarthy	b) Arthur Samuel	c) William S	d) Dennis Ritchie

49 The process of quantifying data is referred to as

a) Structure	b) Decoding	c) Coding	d) Enumeration

50 Linear Regression is the supervised machine learning model in which the model finds the best fit ___ between the independent and dependent variable.

a) Linear line	b) Non linear line	c) Curved line	d) All of the above

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Reg No.:_______________ Name:______________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (S, FE) Examination December 2024 (2019 Scheme)

Course Code: ADT308

Course name: COMPREHENSIVE COURSE WORK

Max. Marks: 50 Duration: 1Hour

Instruction ***(1) Each question carries one mark. No negative marks for wrong answers***
(2) Total number of questions: 50
(3) All questions are to be answered. Each question will be followed by 4 possible answers of
which only ONE is correct.
(4) If more than one option is chosen, it will not be considered for valuation.

1. In classification, what does the term "class label" refer to?
   a) Name of the model b) Output of a regression model c) Predicted category of an input d) Input features of a model

2. What is the role of a confusion matrix in classification?
   a) Visualizing decision boundaries b) Evaluating model performance c) Selecting hyperparameters d) Handling missing data

3. Among the following option identify the one which is not a type of learning
   a) Semi Unsupervised Learning b) Supervised Learning c) Reinforcement Learning d) Unsupervised Learning

4. Which of the following machine learning algorithm is based upon the idea of bagging?
   a) Decision- tree b) Random Forest c) Classification d) Regression

5. What happens when the learning rate is high?
   a) It always reaches the minima quickly b) It reaches the minima very slowly c) It overshoots the minima d) Nothing happens

6. Gradient descent tries to _____________
   a) maximize the cost function b) minimize the cost function c) minimize the learning rate d) maximize the learning rate.

7. Identify the type of learning in which labeled training data is used.

a) Semi Unsupervised Learning
b) Supervised Learning
c) Reinforcement Learning
d) Unsupervised Learning

8. How does the Support Vector Machine (SVM) algorithm work in classification?

a) It finds the nearest neighbors for each data point
b) It fits a linear hyperplane to separate classes
c) It uses decision trees to make predictions
d) It performs clustering based on data density

9. What is the purpose of the epochs parameter in neural network training?

a) The number of layers in the neural network
b) The number of training examples processed in one iteration
c) The learning rate for weight updates
d) The number of complete passes through the entire training dataset

10. What is the concept of entropy in the context of decision trees?

a) The measure of impurity or disorder in a set of data
b) The depth of the decision tree
c) The ratio of training to testing data
d) The number of leaf nodes in the tree

11 Which sorting algorithm is best suited for sorting a nearly sorted list?

a) Quick Sort
b) Merge Sort
c) Insertion Sort
d) Selection Sort

12 In linked lists there are no NULL links in

a) Singly linked list
b) Linear doubly linked list
c) Header linked list
d) circular linked list

13 A graph is a tree if and only if graph

a) is Directed graph
b) Contains no cycles
c) is Planar
d) is completely connected

14 The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree?

a) 2
b) 3
c) 4
d) 5

15 A linear list in which each node has pointers to point to the predecessor and successor nodes is called as

a) Singly Linked List
b) Circular Linked List
c) Doubly Linked List
d) Linear Linked List

16 The matrix contains m rows and n columns. The matrix is called Sparse Matrix if ________

a) Total number of Zero elements > (m*n)/2
b) Total number of Zero elements =m+n
c) Total number of Zero elements = m/n
d) Total number of Zero elements = m-n

17 A data structure in which elements can be inserted or deleted at/from both ends but not in the middle is?

a) Queue b) Circular Queue c) Double ended queue d) Priority queue

18 The postfix form of the expression (A+ B)*(C*D- E)*F / G is?

a) AB+CD*E-*F*G/ b) AB+CD-E**F*G/ c) AB+CD*E-*F/G* d) AB+CD*E/*F*G-

19 Consider a queue implemented using a linear array of size `MAX_SIZE`. If `front = 0` and `rear = MAX_SIZE - 1`, what does this indicate?

a) The queue is full. b) The queue is empty. c) The queue is full, but there is space at the beginning. d) The queue is empty, but there is space at the end.

20 The number of edges in a complete graph of n vertices is

a) n(n+1)/2 b) n(n-1)/2 c) n2 /2 d) n

21 A solution to the problem of external fragmentation is ____________

a) compaction b) larger memory space c) smaller memory space d) none of the mentioned

22 What are the two kinds of semaphores?

a) mutex & counting b) binary & counting c) counting & decimal d) decimal & binary

23 In the ______ algorithm, the disk arm starts at one end of the disk and moves toward the other end, servicing requests till the other end of the disk. At the other end, the direction is reversed and servicing continues.

a) SCAN b) FCFS c) C-SCAN d) None of the above

24 What is Response time?

a) the total time taken from the submission time till the completion time b) the total time taken from the submission time till the first response is produced c) the total time taken from submission time till the response is output d) none of the mentioned

25 The real difficulty with Shortest Job First(SJF) in short term scheduling is __________

a) it is cyclic b) knowing the length of the next CPU request c) It's not flexible d) it is too complex to understand

26 What is 'Aging'?

a) keeping track of cache contents b) keeping track of what pages are currently residing in memory c) keeping track of how many times a given page is referenced d) increasing the priority of jobs to ensure termination in a finite time

27 With round robin scheduling algorithm in a time shared system ____________

a) using very large time slices converts it into First come First served scheduling algorithm
b) using very small time slices converts it into First come First served scheduling algorithm
c) using extremely small time slices increases performance
d) using very small time slices converts it into Shortest Job First algorithm

28 The interval from the time of submission of a process to the time of completion is termed as ____________

a) waiting time
b) turnaround time
c) response time
d) throughput

29 In internal fragmentation, memory is internal to a partition and ____________

a) is being used
b) is not being used
c) is always used
d) none of the mentioned

30 In fixed size partition, the degree of multiprogramming is bounded by ____________

a) the number of partitions
b) the CPU utilization
c) the memory size
d) all of the mentioned

31 The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to ____________________ integrity constraint.

a) Referential
b) Primary
c) Referencing
d) Specific

32 The ability to query data, as well as insert, delete, and alter tuples, is offered by ____________

a) TCL (Transaction Control Language)
b) DCL (Data Control Language)
c) DDL (Data Definition Langauge)
d) DML (Data Manipulation Langauge)

33 After groups have been established, SQL applies predicates in the ____________ clause, allowing aggregate functions to be used.

a) Where
b) Having
c) Group by
d) With

34 For designing a normal RDBMS which of the following normal form is considered adequate?

a) 4NF
b) 3NF
c) 2NF
d) 1NF

35 The logical design, and the snapshot of the data at a given instant in time is known as?

a) Instance & Relation
b) Relation & Schema
c) Domain & Schema
d) Schema & Instance

36 What does the FROM clause in an SQL query represent?

a) Attributes to be retrieved
b) Conditions for selection
c) Relations to be scanned
d) Grouping criteria

37 Based on the given example, what is the purpose of the condition Dnumber = Dno in the WHERE clause?

a) Selection condition
b) Join condition
c) Grouping condition
d) Sorting condition

38 In the context of normalization, what does the term "functional dependency" mean?

a) A relationship between primary and foreign keys.
b). A relationship between two or more attributes in a table.
c) The unique identifier for a table.
d) The process of removing duplicates from a table.

39 What is the key difference between Second Normal Form (2NF) and Third Normal Form (3NF)?

a) 2NF allows partial dependencies, while 3NF does not.
b) 2NF requires a composite primary key, while 3NF does not.
c) 3NF requires a composite primary key, while 2NF does not.
D) 3NF allows transitive dependencies, while 2NF does not.

40 Consider the following action:
TRANSACTION.....
Commit;
ROLLBACK;
What does Rollback do?

a) Undoes the transactions before commit
b) Clears all transactions
c) Redoes the transactions before commit
d) No action

41 In supervised learning, which of the following is required?

a) Labeled data
b) Only numerical data
c) Unlabeled data
d) Only categorical data

42 Which of the following is a common method for data preprocessing?

a) Data normalization
b) Data storage
c) Data aggregation
d) Data visualization

43 What technique is commonly used in data science for making predictions?

a) Data cleaning
b) Data Storage
c) Machine Learning
d) Data Encoding

44 What type of data is considered unstructured?

a) Data in relational databases
b) Data in spreadsheets
c) Data in CSV files
d) Text documents and images

45 Which of the following input can be accepted by DataFrame?

a) DataFrame
b) Series
c) Structured ndarray
d) All of the mentioned

46 Which of the following is used to extract data from HTML code of websites?

a) Webscraping
b) Webcleaning
c) Webdredging
d) All of the mentioned

47 What does K stand for in K means algorithm?

a) number of clusters
b) number of data
c) number of attributes
d) number of iterations

48 Which of the following is not a supervised learning

a) PCA b) Naïve Bayessian c) Linear regression d) Decision Tree

49 Choose a disadvantage of decision trees among the following.

a) Decision trees are robust to outliers b) Factor analysis c) Decision trees are prone to overfit d) All of the above

50 Choose the Python data structure responsible for the storage and manipulation of tabular data in Data Science

a) Array b) List c) Dictionary d) DataFrame

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Reg No.:_______________ Name:_______________________

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Sixth Semester B.Tech Degree (R, S) Examination May 2024 (2019 Scheme)

Course Code: ADT308

Course name: COMPREHENSIVE COURSE WORK

Max. Marks: 50 Duration: 1Hour

(1) Each question carries one mark. No negative marks for wrong answers
(2) Total number of questions: 50
(3) All questions are to be answered. Each question will be followed by 4 possible answers of
which only ONE is correct.
(4) If more than one option is chosen, it will not be considered for valuation.

1 Which of the following is NOT a valid deadlock prevention scheme?

a) Release all resources before requesting a new resource
b) Number the resources uniquely and never request a lower numbered resource than the last one requested.
c) Never request a resource after releasing any resource
d) Request and all required resources be allocated before execution.

2 Consider a virtual memory system with FIFO page replacement policy. For an arbitrary page access pattern, increasing the number of page frames in main memory will

a) Always decrease the number of page faults
b) Always increase the number of page faults
c) Some times increase the number of page faults
d) Never affect the number of page faults

3 Suppose the time to service a page fault is on the average 10 milliseconds, while a memory access takes 1 microsecond. Then a 99.99% hit ratio results in average memory access time of

a) 1.9999 milliseconds
b) 1 millisecond
c) 9.999 microseconds
d) 1.9999 microseconds

4 Which of the following scheduling algorithms is non-preemptive?

a) Round Robin
b) First-In First-Out
c) Multilevel Queue Scheduling
d) Multilevel Queue Scheduling with Feedback

13 A standard circular Queue 'q' implemented whose size is 11 from 0 to 10. The front & rear pointers start to point at q[2]. The ninth element be added at which position?

a) q[0] b) q[1] c) q[9] d) q[10]

14 The preorder traversal of a binary search tree is 15, 10, 12, 11, 20, 18, 16, 19. Which one of the following is the postorder traversal of the tree?

a) 11,12,10,16,19,18,20,15 b) 20,19,18,16,15,12,11,10 c) 10,11,12,15,16,18,19,20 d) 19,16,18,20,11,12,10,15

15 A hash table contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is key % 10. If the values 43, 165, 62, 123, 142 are inserted in the table, in what location would the key value 142 be inserted?

a) 2 b) 3 c) 4 d) 6

16 Let P be a singly linked list, Let Q be the pointer to an intermediate node x in the list.What is the worst-case time complexity of the best known algorithm to delete the node x from the list?

a) O(n) b) O(n log n) c) O(1) d) None of these

17 The result evaluating the postfix expression is: 10 5 + 60 6 / * 8 -

a) 284 b) 213 c) 142 d) 71

18 The best data structure to check whether an arithmetic expression has balanced parentheses is a

a) Stack b) Queue c) Tree d) List

19 In a depth-first traversal of a graph G with n vertices,k edges are marked as tree edges. The number of connected components in G is

a) K b) k+1 c) n-k d) n-k-1

20 What is the maximum number of swaps that can be performed in the Selection Sort algorithm?

a) n-1 b) n c) 1 d) n-2

21 Which one of the options given below refers to the degree of a relation in relational database systems?

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20 What is the maximum number of swaps that can be performed in the Selection Sort algorithm?

a) n-1 b) n c) 1 d) n-2

21 Which one of the options given below refers to the degree of a relation in relational database systems?

a) Number of attributes of its relation schema.
b) Number of tuples stored in the relation.
c) Number of entries in the relation.
d) Number of distinct domains of its relation schema.

22 Which of the following statements is/are TRUE with respect to deadlocks?

a) Circular wait is a necessary condition for the formation of deadlock
b) In a system where each resource has more than one instance, a cycle in its wait-for graph indicates the presence of a deadlock
c) If the current allocation of resources to processes leads the system to unsafe state, then deadlock will necessarily occur
d) None of these

23 In SQL, which of the following is an aggregate function?

a) Avg
b) Select
c) Ordered by
d) Distinct

24 Which of the following is generally used for performing tasks like creating the structure of the relations, deleting relation?

a) DML
b) Query
c) Relational Schema
d) DDL

25 Which one of the following given statements possibly contains the error?

a) select * from emp where empid = 10003;
b) select empid from emp where empid = 10006;
c) select empid from emp;
d) select empid where empid = 1009 and Lastname = 'GELLER';

26 A Database Management System is a type of _________software.

a) It is a type of system software
b) It is a kind ~~of~~ application software
c) It is a kind of general software
d) None of these

27 Which of the following is a top-down approach in which the entity's higher level can be divided into two lower sub-entities?

a) Aggregation
b) Generalization
c) Specialization
d) None of these

28 Which one of the following refers to the total view of the database content?

a) Conceptual view
b) Physical view
c) Internal view
d) None of these

29 Which of the following statements is correct?
1. A self-join may be implemented in SQL using table aliases.
2. The outer-join operation is a fundamental relational algebra procedure.
3. The procedures of natural join and outside join are the same.

a) 1 is correct
b) 1 and 2 is correct
c) 2 and 3 is correct
d) None of these

30 Location transparency in distributed databases allows database users, programmers, and administrators to handle data as if it were stored in a single location. A SQL query with location transparency must provide the following information:

a) Fragments
b) Locations
c) Local formats
d) Inheritances

31 Which algorithm is used for finding frequent itemsets in transactional databases?

a) Decision Trees
b) K-Means clustering
c) Apriori algorithm
d) Support Vector Machines (SVM)

32 In the context of machine learning, what is the term "bias-variance trade-off" referring to?

a) The trade-off between the quality of training data and testing data
b) The trade-off between the complexity of a model and its ability to generalize
c) The trade-off between the number of features and the size of the dataset
d) The trade-off between precision and recall in classification

33 Which algorithm is used for hierarchical clustering?

a) K-Means clustering
b) Agglomerative clustering
c) DBSCAN
d) PCA

34 Which method can be used to handle missing data in a dataset?

a) Removing the entire column with missing data
b) Replacing missing data with the median value of the column
c) Ignoring the rows with missing data during analysis
d) All of the above

35 Which technique is used to combat the vanishing gradient problem in deep neural networks?

a) ReLU activation function
b) Sigmoid activation function
c) Tanh activation function
d) Batch normalization

36 In the context of Support Vector Machines (SVM), what is the "kernel trick"?

a) A technique to add more features to the dataset b) A method to increase the regularization term c) A way to perform dimensionality reduction d) A way to implicitly map data to higher-dimensional spaces

37 What is the purpose of regularization in linear regression?

a) To make the model more complex b) To avoid underfitting c) To encourage overfitting d) To reduce the complexity of the model

38 K-Nearest Neighbors (KNN) is classified as what type of machine learning algorithm?

a) Instance-based learning b) Parametric learning c) Non-parametric learning d) Model-based learning

39 Which of the following is not a supervised machine learning algorithm?

a) K-means b) Naive Bayes c) SVM d) Decision Tree

40 Which of the following statements is not true about boosting?

a) It mainly increases the bias and the variance b) It mainly decreases the bias and the variance c) It is a technique for solving two-class classification problems d) None of these

41 Ensemble learning is a machine learning technique that involves:

a) Training multiple models and combining their predictions b) Training a single model on a large dataset c) Training a model using a deep neural network d) Training a model using reinforcement learning

42 Which ensemble learning algorithm assigns weights to base models based on their performance?

a) AdaBoost b) Random Forest c) Gradient Boosting d) Stacking

43 Which of the following statement is not true about Naïve Bayes classifier algorithm?

a) It cannot be used for Binary as well as multi-class classifications
b) It is the most popular choice for text classification problems
c) It performs well in Multi-class prediction as compared to other algorithms
d) It is one of the fast and easy machine learning algorithms to predict a class of test datasets

44 Which one of the following models is a generative model used in machine learning?

a) Linear Regression
b) Logistic Regression
c) Naïve Bayes
d) Support vector machines

45 Which of the following can be considered as the classification or mapping of a set or class with some predefined group or classes?

a) Data set
b) Data Characterization
c) Data Sub Structure
d) Data Discrimination

46 In order to integrate heterogeneous databases, how many types of approaches are there in the data warehousing?

a) 1
b) 2
c) 3
d) 4

47 Which of the following is generally used by the E-R model to represent the weak entities?

a) Diamond
b) Doubly outlined rectangle
c) Dotted rectangle
d) Both B & C

48 Which of the following is finally produced by Hierarchical Clustering?

a) Final estimate of cluster centroids
b) Tree showing how close things are to each other
c) Assignment of each point to clusters
d) All of the above

49 Which clustering algorithm is suitable for high-dimensional data?

a) K-Means
b) DBSCAN
c) Agglomerative
d) Mean-Shift

50 Which of the following statements is not true about Random forests?

a) It has high complexity
b) Construction of Random forests are much easier than decision trees
c) Construction of Random forests are time-consuming than decision trees
d) More computational resources are required to implement Random Forest algorithm

*****